Phase cluster sets and refined terminal selection in critical raw gradient descent

Abstract

For an exact three-variable gradient-descent map, we refine a uniform square-root selection law by computing its complete second-order correction with a uniform cubic remainder. A second coordinate calculation provides an exact algebraic certificate for the coefficients. For each fixed small positive tangential displacement, an explicit rescaling construction produces the outgoing unstable parametrization of the amplitude map. Its terminal-coordinate function identifies the punctured cluster set exactly as the image of a continuous logarithmic-phase function. The punctured limit exists if and only if that function is constant. **Constancy and nonconstancy remain unresolved.** The result asserts neither existence nor failure of the punctured limit.

Keywords: critical gradient descent; terminal selection; second-order correction; logarithmic phase; cluster sets; unstable parametrization

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1 Map, scope, and statements

Set

$$u = \frac{10}{3}, \quad h = \frac{1}{3000}, \quad A = \sqrt{\frac{3}{5}}, \quad C = A^{-1}, \quad \lambda = \frac{9}{68}.$$

The objective and exact simultaneous gradient-descent update with stepsize $1/30$ are

$$\begin{gathered} f(a,b,c) = 50a^2b^2 + \frac{(ac-1)^2}{200}, \\ a^+ = a - uab^2 - hc(ac-1), \qquad b^+ = (1-ua^2)b, \\ c^+ = c - ha(ac-1). \end{gathered} \tag{1}$$

Every right-hand side is evaluated at the old state. There is no clipping, projection, cutoff, or numerical modification. The initial family is

$$(a_0, b_0, c_0) = (A+s, b, (A+s)^{-1}). \tag{2}$$

This report extends the uniform local analysis in the companion report [2]. Its motivating optimization context is the final NeurIPS 2025 paper of Liu, Zhang, Du and Zhao [1]. We analyze only the specified raw map and family. No originality claim, original-paper-error claim, or identification with a source-paper experimental seed is made.

Theorem 1 (Refined uniform terminal selection)**.** *There are $\delta > 0$ and $M < \infty$ such that, whenever $|s| + |b| < \delta$ and $b \neq 0$, the orbit is captured in an $O(\sqrt{s^2 + b^2})$ neighborhood of $(A, 0, C)$, b_n never vanishes and alternates signs, and*

$$(a_n, b_n, c_n) \longrightarrow (a_\infty, 0, a_\infty^{-1}), \qquad 0 < a_\infty < A.$$

Convergence is geometric after a possibly parameter-dependent transient. Define

$$\rho = \sqrt{s^2 + \lambda b^2}, \qquad \chi = -\frac{999913\sqrt{15}}{153}, \qquad \psi = -\frac{374949\sqrt{15}}{289}. \tag{3}$$

Then, uniformly over these parameters,

$$\left| a_\infty - A + \rho + \frac{\chi(s^3 + \rho^3) + \psi s b^2}{2\rho} \right| \leq M\rho^3. \tag{4}$$

There is no lower bound on $|b|/|s|$.

In the capture neighborhood $ua^2 - 1 > 0$, so $v = |b|$ evolves under the actual amplitude map

$$G(a, v, c) = \big(a - uav^2 - hc(ac - 1),\ (ua^2 - 1)v,\ c - ha(ac - 1)\big). \tag{5}$$

Thus this reduction is exact, rather than a continuous-time approximation.

Theorem 2 (Exact phase cluster set)**.** *Fix any sufficiently small $s > 0$, and set*

$$\mu_s = u(A + s)^2 - 1 > 1, \qquad I_s(x) = (A + s, x, (A + s)^{-1}).$$

There are $\eta_s > 0$ and a continuous outgoing parametrization P_s, defined at least for $0 \leq x \leq \mu_s^2 \eta_s$, such that

$$P_s(x) = \lim_{n \to \infty} G^n(I_s(\mu_s^{-n} x)), \tag{6}$$

$$P_s(x) = I_s(x) + O_s(x^2), \qquad G(P_s(x)) = P_s(\mu_s x), \tag{7}$$

where the equivariance identity is asserted when both parameters lie in the domain. Every $P_s(x)$ with $x > 0$ converges to a strictly subcritical fixed point. Let E denote its terminal first coordinate. The function

$$\sigma_s(\tau) = E(P_s(\eta_s \mu_s^\tau)), \qquad 0 \leq \tau \leq 1, \tag{8}$$

is continuous and satisfies $\sigma_s(0) = \sigma_s(1)$. The cluster set of $a_\infty(s, b)$ as nonzero $b \to 0$ is exactly

$$\sigma_s([0, 1]) = [\min \sigma_s, \max \sigma_s]. \tag{9}$$

More precisely, choose $n = n(b)$ so that $x(b) = \mu_s^n |b| \in [\eta_s, \mu_s \eta_s)$. Then

$$a_\infty(s, b) - E(P_s(x(b))) \longrightarrow 0. \tag{10}$$

Consequently, the punctured limit exists if and only if σ_s is constant, and in that case equals the constant.

Corollary 1.1. *With one constant M independent of sufficiently small $s > 0$, every punctured cluster value ℓ satisfies*

$$\left|\ell - \left(A - s + \frac{999913\sqrt{15}}{153}s^2\right)\right| \le Ms^3. \tag{11}$$

In particular, the cluster-set diameter is at most $2Ms^3$.

Unresolved issue. Neither constancy nor nonconstancy of σ_s is proved here for any fixed sufficiently small $s > 0$. The center in (11) is an enclosure center, not an asserted exact limit. All neighborhoods are existential; the large coefficient does not provide a numerically usable smallness threshold.

2 Coordinates and uniform capture

We give the estimates needed below, including a proof of the capture input of [2]. All un-subscripted constants are independent of (s, b) and time. In a positive neighborhood define

$$r = ac - 1, \quad d(p) = p^2 - p^{-2} = a^2 - c^2, \quad t = p - A, \quad v = |b_n|, \quad D(p) = p^2 + p^{-2}.$$

These are smooth coordinates: $d'(p) = 2p + 2p^{-3} > 0$, and the Jacobian of $(a, c) \mapsto (a^2 - c^2, ac - 1)$ is $2(a^2 + c^2) > 0$. At fixed p,

$$a = p + \frac{p^{-1}}{D(p)}r + O(r^2), \qquad c = p^{-1} + \frac{p}{D(p)}r + O(r^2). \tag{12}$$

Direct expansion of (1) gives

$$r^+ = Qr - uacv^2, \tag{13}$$

$$Q = 1 - h(a^2 + c^2) + uha^2v^2 + h^2acr,$$

$$d(p^+) - d(p) = -2ua^2v^2 + u^2a^2v^4 + 2uhacrv^2 - h^2(a^2 - c^2)r^2. \tag{14}$$

The absence of a term linear in r in (14) is essential. Initially $t = s$, $r = 0$, and $v = |b|$.

Put $D = D(A) = 34/15$ and $q_0 = 1 - hD \in (0, 1)$. Fix $q_0 < q < 1$ and $\sqrt{q} < m < 1$. In a sufficiently small working neighborhood U,

$$0 < Q \le q, \qquad ua^2 - 1 \ge m, \qquad 0 < uac \le D_0.$$

Choose $L \ge D_0/(m^2 - q)$. The cone

$$-Lv^2 \le r \le 0 \tag{15}$$

propagates for every step justified in U, since

$$r^+ \le 0, \qquad |r^+| \le (qL + D_0)v^2 \le L(v^+)^2.$$

Use a larger chart containing the image of U, so that one-step coordinate estimates do not presuppose capture of that image. On the cone, smooth inversion of (14) gives

$$t^+ - t = -k(p)v^2 + O(v^4), \qquad k(p) = \frac{up^5}{p^4+1}. \tag{16}$$

After reducing U, fixed positive k_*, k^* satisfy

$$-k^* v^2 \le t^+ - t \le -k_* v^2.$$

Set $K = k(A)$, $\kappa = 2uA$, so $K/\kappa = \lambda$. Uniformly,

$$t^+ - t = -Kv^2 + O(|t|v^2 + v^4), \qquad v^+ = v[1 + \kappa t + O(t^2 + v^2)]. \tag{17}$$

For $H = t^2 + \lambda v^2$, cubic terms cancel, yielding

$$|H^+ - H| \le M_0 H v^2. \tag{18}$$

Here is a closed induction establishing capture. Suppose $H_j \le 2\rho^2$ and the cone hold through time n. The last step is legitimate in the larger chart and

$$t_{n+1} \ge -\sqrt{2}\rho - \frac{2k^*}{\lambda}\rho^2, \qquad k_* \sum_{j=0}^{n} v_j^2 \le s - t_{n+1} \le B\rho$$

for a fixed B, after a fixed smallness reduction. Therefore

$$H_{n+1} \le \rho^2 \exp(M_0 B\rho/k_*) \le \tfrac{3}{2}\rho^2 < 2\rho^2.$$

The propagated cone and this improved bound return the image to U when ρ is sufficiently small. Since $H_0 = \rho^2$, induction closes and gives

$$|t_n| \le \sqrt{2}\rho, \quad v_n^2 \le 2\rho^2/\lambda, \quad |r_n| \le 2L\rho^2/\lambda, \quad \sum_{n\ge 0} v_n^2 \le C_0\rho. \tag{19}$$

The final sum follows by telescoping the monotone bounded t_n. Summing (18) also gives, at every finite time,

$$|H_n - \rho^2| \le C_1\rho^3. \tag{20}$$

The sum and cone imply $v_n, r_n \to 0$, while $t_n \downarrow t_\infty$. Thus the original state converges to $(A + t_\infty, 0, (A + t_\infty)^{-1})$, and

$$|t_\infty^2 - \rho^2| \le C_1\rho^3.$$

For small $\rho > 0$, $|t_\infty| \ge \rho/\sqrt{2}$. A positive t_∞ would make $ua_n^2 - 1 > 1$ by a fixed amount eventually, contradicting $v_n \to 0$ and $v_n > 0$. Hence

$$t_\infty < 0, \qquad t_\infty = -\rho + O(\rho^2). \tag{21}$$

Nonvanishing and sign alternation follow from $ua_n^2 - 1 \ge m$. At late times this multiplier is bounded above by some $\theta < 1$. Then v_n decays geometrically, and the cone and drift tail sum prove geometric convergence of a_n, c_n as well. The constants and transient may depend on the initial point. This proves the capture and convergence assertions without requiring a bound on the residence time near the unstable fixed line.

3 Refined terminal expansion

3.1 The initial stable-coordinate lag

Define

$$R_0 = -\frac{u}{hD}, \qquad e = r - R_0 v^2.$$

Since $q_0 R_0 - u = R_0$, equations (13) and (17), on a captured orbit, imply

$$e^+ = q_0 e + O(\rho v^2). \tag{22}$$

Here $e_0 = -R_0 b^2$ is retained explicitly. Because $v^+ \geq mv$ and $q_0 < m^2$, iteration gives

$$|e_n| \leq |R_0| b^2 q_0^n + C_2 \rho v_n^2. \tag{23}$$

Indeed the forced sum is bounded by

$$C\rho \sum_{j<n} q_0^{n-1-j} v_j^2 \leq C\rho v_n^2 m^{-2} \sum_{k<n} (q_0/m^2)^k.$$

Using (19), we obtain

$$\sum_{n\geq 0} |e_n| v_n^2 \leq C b^2 \rho^2 \sum_{n\geq 0} q_0^n + C\rho \sum_{n\geq 0} v_n^4 \leq C_3 \rho^4. \tag{24}$$

This bound includes arbitrarily small nonzero b and makes no center-manifold initialization assumption.

3.2 Cubic correction and total error

Let $k_1 = k'(A)$, $B = 2u/D$, and $\beta = BR_0 = -2u^2/(hD^2)$. Taylor expansion of the exact identities gives

$$t^+ - t = -Kv^2 - k_1 t v^2 + O(\rho^2 v^2), \tag{25}$$

$$v^+ - v = \kappa t v + u t^2 v + \beta v^3 + Bev + O(\rho^3 v). \tag{26}$$

For (26), expand $a^2 = p^2 + 2r/D(p) + O(r^2)$ and substitute $r = R_0 v^2 + e$; all neglected terms obey the stated bound since $|e| \leq Cv^2$. Let

$$\alpha = -2k_1 + \lambda(\kappa^2 + 2u), \qquad \gamma = K^2 + 2\lambda\beta.$$

Squaring (25)–(26), including their discrete quadratic increments, yields

$$H^+ - H = \alpha t^2 v^2 + \gamma v^4 + 2\lambda B e v^2 + O(\rho^3 v^2). \tag{27}$$

Choose

$$\psi = \gamma/K, \qquad \chi = (\alpha + 2\kappa\psi)/(3K), \qquad \mathcal{I} = t^2 + \lambda v^2 + \chi t^3 + \psi t v^2. \tag{28}$$

The leading change of the cubic correction is

$$(-3\chi K + 2\psi\kappa)t^2v^2 - \psi K v^4 + O(\rho^3 v^2).$$

Thus both quartic terms in (27) cancel and

$$|\mathcal{I}^+ - \mathcal{I}| \le C\rho^3 v^2 + C|e|v^2.$$

Equations (19) and (24) give the controlled accumulated error

$$t_\infty^2 + \chi t_\infty^3 = \rho^2 + \chi s^3 + \psi s b^2 + O(\rho^4). \tag{29}$$

This is an approximate invariant, not an exact conservation law. From (21), write $t_\infty = -\rho + z$ with $z = O(\rho^2)$. Then the left side of (29) is

$$\rho^2 - 2\rho z - \chi\rho^3 + O(\rho^4).$$

Division by $2\rho > 0$ gives exactly (4). For transparency, the constants are

$$\begin{gathered} D = \frac{34}{15}, \quad K = \frac{3\sqrt{15}}{17}, \quad \kappa = \frac{4\sqrt{15}}{3}, \quad k_1 = \frac{1005}{289}, \\ \beta = -\frac{3750000}{289}, \quad \psi = -\frac{374949\sqrt{15}}{289}, \quad \chi = -\frac{999913\sqrt{15}}{153}. \end{gathered} \tag{30}$$

They follow from $A^2 = 3/5$ and (28); none is fitted. This proves Theorem 1.

4 Independent coefficient certificate

A different coordinate calculation verifies the coefficients without using the p-coordinate correction. In this section only, write $z = a - A$, $w = b^2$, and $r = ac - 1$, so $c = (1 + r)/(A + z)$. The exact next coordinates are

$$\begin{aligned} Z &= z - u(A+z)w - hcr, \\ W &= [u(A+z)^2 - 1]^2 w, \\ R &= [1 - h((A+z)^2 + c^2)]r + h^2(A+z)cr^2 \\ &\qquad - u(A+z)cw + uh(A+z)^2 rw. \end{aligned}$$

Give z, w, r weights $1, 2, 2$, respectively, and form

$$\mathcal{J} = z^2 + \lambda w + \chi z^3 + \psi zw + kzr + lz^2 r + mwr + nr^2. \tag{31}$$

Solving for vanishing weighted-degree at most four in $\mathcal{J}(Z, W, R) - \mathcal{J}(z, w, r)$ gives the same χ, ψ as (3), together with

$$k = -\frac{5\sqrt{15}}{17}, \quad l = \frac{24997875}{578}, \quad m = \frac{2249775}{578}, \quad n = \frac{375}{1156}. \tag{32}$$

The portable exact-arithmetic script `check_coefficients.py` derives these coefficients by solving the polynomial equations, rather than substituting the desired answer into them. It also checks uniqueness of the solution and zero residual after substitution.

To see why this is a genuine uniform certificate, every $(z, 0, 0)$ is fixed, so the analytic defect has no term independent of (w, r). The weighted-degree cancellation therefore implies

$$|\mathcal{J}^+ - \mathcal{J}| \le C\rho^3(w + |r|)$$

whenever $|z| \le C\rho$ and $w + |r| \le C\rho^2$. These bounds hold by capture and the cone. Moreover, $r \le 0$ and

$$c^+ - c = ha|r|.$$

Since a is bounded below and c is bounded with total variation $O(\rho)$, this gives $\sum |r_n| = O(\rho)$. Alternatively the cone and (19) give that sum directly. Also $\sum w_n = O(\rho)$. Thus the total defect is $O(\rho^4)$.

Initially (31) equals $s^2+\lambda b^2+\chi s^3+\psi s b^2$, and at the terminal point it equals $(a_\infty - A)^2 + \chi(a_\infty - A)^3$. Hence this separate calculation reproduces (29) and the refined expansion. The symbolic certificate checks finite algebra; the preceding analytic summation is what justifies the infinite-time conclusion.

5 Construction of the outgoing parametrization

Fix $s > 0$ in the established range. Write $\mu = \mu_s$, $z_s = (A + s, 0, (A + s)^{-1})$, and $I = I_s$. In the smooth (p, v, r) coordinates, the derivative of the amplitude map at z_s is diagonal, with eigenvalues

$$1, \qquad \mu > 1, \qquad q_s = 1 - h[(A + s)^2 + (A + s)^{-2}] \in (0, 1).$$

The first assertion follows from (14); the other two follow from (5) and (13). Choose a convex coordinate neighborhood V and a norm in which the Lipschitz constant L_s of G satisfies

$$\mu < L_s < \mu^2. \tag{33}$$

Continuity of the derivative makes this possible for each fixed $s > 0$. No uniformity as $s \downarrow 0$ is claimed here. In these coordinates,

$$\|G(I(x)) - I(\mu x)\| \le C_s x^2. \tag{34}$$

For sufficiently small $x_0 > 0$ define, on $0 \le x \le x_0$,

$$P_n(x) = G^n(I(\mu^{-n}x)).$$

These iterates remain in V. Indeed, comparison at step j with $I(\mu^{j-n}x)$, using (33)–(34), bounds the error by

$$C_s x^2 \mu^{2(j-n)} \sum_{k=0}^{j-1} L_s^k \mu^{-2(k+1)} \le C_s' x^2 \mu^{2(j-n)}.$$

Choose x_0 so that the reference segment and this error lie strictly inside V. Induction then justifies every comparison, rather than assuming local retention.

The initial discrepancy between P_{n+1} and P_n is the one-step defect at $I(\mu^{-n-1}x)$, followed by n common iterates. Consequently

$$\sup_{0\le x\le x_0} \|P_{n+1}(x) - P_n(x)\| \le C_s'' x_0^2 (L_s/\mu^2)^n. \tag{35}$$

The ratio is strictly less than one, so P_n converges uniformly to a continuous P_s satisfying (6) and $P_s(x) = I_s(x) + O_s(x^2)$. Passing to the limit in $G(P_n(x)) = P_{n+1}(\mu x)$ proves equivariance. Moreover $P_s(0) = z_s$ and $P_s(\mu^{-n}x) \to z_s$. Reduce x_0 so that the amplitude coordinate is positive for $x > 0$, and choose $\eta_s > 0$ with $\mu^2\eta_s \le x_0$. This is a parametrization of the actual amplitude map; no abstract unstable-manifold theorem or flow embedding is needed.

6 Strict terminal selection and endpoint continuity

6.1 Forward convergence from the outgoing curve

For $x > 0$, $P_n(x)$ lies on an initial-family orbit with initial amplitude $\mu^{-n}x$. Every fixed additional forward iterate inherits, upon letting $n \to \infty$, the uniform finite-time bounds

$$-Lv^2 \le r \le 0, \qquad H \le 2s^2, \qquad |H - s^2| \le C_1 s^3. \tag{36}$$

Indeed its initial scale $\sqrt{s^2 + \lambda\mu^{-2n}x^2}$ tends to s. The drift bound and boundedness imply $\sum v_j^2 < \infty$ on the forward orbit from $P_s(x)$. The cone implies $r_j \to 0$ and p_j converges, so the point converges to a fixed point. Its terminal displacement T obeys

$$|T^2 - s^2| \le C_1 s^3.$$

For sufficiently small $s > 0$ this excludes $T = 0$. Positivity and nonvanishing of the amplitude exclude $T > 0$ by the multiplier argument used above. Hence $T < 0$ and the terminal fixed point is strictly subcritical.

6.2 Continuity of the terminal-coordinate function

Continuity of the parametrization alone does not justify continuity of its terminal limit. We establish the latter locally. Near any subcritical fixed point $(a_*, 0, a_*^{-1})$ sufficiently close

to $(A, 0, C)$, the transverse derivative in (v, r) has diagonal entries

$$ua_*^2 - 1 \in (0, 1), \qquad 1 - h(a_*^2 + a_*^{-2}) \in (0, 1).$$

Choose a compact p interval strictly below A and shrink the transverse radius. Equations (13)–(14) then give throughout a suitable tube

$$|v^+| + |r^+| \le \theta(|v| + |r|), \qquad |p^+ - p| \le C(|v|^2 + |r|^2), \qquad \theta < 1. \tag{37}$$

The quadratic forcing in r is absorbed into the strict transverse contraction gap. The second bound uses the cancellation of the linear residual term in (14). No cone assumption is required for this tube.

Start p in a smaller interior interval and choose the initial transverse radius so that

$$C \sum_{j \ge 0} \theta^{2j} (|v| + |r|)^2$$

is less than the margin to the interval boundary. The estimates close by induction, guarantee retention in the tube, and give a uniform convergent tail for p. The terminal first coordinate E is therefore continuous there, as a uniform limit of continuous finite iterates; at the limit $a = p$.

Each positive-parameter $P_s(x)$ eventually enters the interior of such a tube. Pulling back that interior through a finite prefix gives an open neighborhood of $P_s(x)$ where E exists and is continuous. Also $E(Gz) = E(z)$ where both are defined. The compact fundamental arc

$$\mathcal{K}_s = P_s([\eta_s, \mu\eta_s])$$

therefore has an open neighborhood of continuity. A finite cover by smaller relatively compact neighborhoods implies uniform continuity for points sufficiently near $\mathcal{K}_s$. We have not assumed continuity of E at the original unstable fixed point z_s.

7 Exact cluster set and the remaining question

For sufficiently small $b \neq 0$, the unique integer $n = n(b) \ge 0$ with

$$x(b) = \mu^n |b| \in [\eta_s, \mu\eta_s)$$

tends to infinity as $b \to 0$. Its exact amplitude iterate is $P_n(x(b))$. Uniform convergence (35) implies

$$\|P_n(x(b)) - P_s(x(b))\| \longrightarrow 0.$$

Uniform continuity of E near $\mathcal{K}_s$, proved above, now yields (10). Every terminal cluster value therefore belongs to the compact set $E(\mathcal{K}_s)$.

Conversely, for each $x \in [\eta_s, \mu\eta_s]$, take $b_n = \mu^{-n}x$. Then

$$G^n(I_s(b_n)) = P_n(x) \longrightarrow P_s(x), \qquad a_\infty(s, b_n) \longrightarrow E(P_s(x)).$$

This recovers every value in $E(\mathcal{K}_s)$. The sign of b does not affect the amplitude dynamics or its terminal first coordinate. Thus the two-sided punctured cluster set is exactly $E(\mathcal{K}_s)$.

The function (8) is continuous, and

$$\sigma_s(1) = E(P_s(\mu\eta_s)) = E(G(P_s(\eta_s))) = E(P_s(\eta_s)) = \sigma_s(0).$$

Its image is a compact interval by the intermediate value theorem. This proves (9); compactness together with (10) proves the stated if-and-only-if limit criterion, including the existence direction when the image is a singleton. Theorem 2 follows.

Finally, take $b \to 0$ at fixed $s > 0$ in the uniform bound (4). Since $\rho \to s$ and the correction tends to χs^2, each cluster value satisfies

$$\ell = A - s - \chi s^2 + O(s^3)$$

with the same constant for all sufficiently small s. This proves Corollary 1.1.

What is not concluded. Discrete equivariance supplies periodicity in logarithmic phase, not invariance under arbitrary real time. It permits a nonconstant continuous phase function but does not prove one. Neither the enclosure nor its shrinking width proves that the cluster interval is a singleton at fixed positive s. Establishing constancy or nonconstancy of σ_s for the specified raw map remains the precise open question. All estimates require a sufficiently small existential neighborhood chosen after their constants; no practical numerical threshold or finite-precision conclusion is supplied.

A Reproduction and certificate scope

With a standard LaTeX distribution, run `bash build.sh` to compile this report twice for cross-references. With Python and SymPy, run

```
python3 check_coefficients.py
```

The script expands the exact map in weighted degree and solves for the six unknown coefficients in (31). It asserts that the solution is unique, agrees with (3) and (32), and annihilates every retained coefficient. This exact finite algebra does not replace capture, summability, endpoint continuity, or the cluster-set argument, all of which are supplied above.

References

[1] Liming Liu, Zixuan Zhang, Simon Shaolei Du and Tuo Zhao. *A Minimalist Example of Edge-of-Stability and Progressive Sharpening*. Advances in Neural Information Processing Systems, 2025, final proceedings version. `https://papers.nips.cc/paper_files/paper/2025/file/15eac388458e8a577f96edf7a40d0cbc-Paper-Conference.pdf`.

[2] *Uniform two-parameter selection in critical raw gradient descent*. Companion mathematical report 12, 2026. Supplied with this report series.